Om Prakash Choudhary

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About Me

I am passionate about **generative modeling**, with hands-on experience in both **text** and **image** generation. I actively stay up to date with the latest research developments in the field.

Education

Indian Institute of Science, *B.Tech in Mathematics and Computing (3rd Year)* — Aug 2023 - Present

- GPA: 8.5 / 10.0
- **Relevant Courses:** Machine Learning, Optimization Theory, Multi-Armed Bandits, Algorithms, Computer Systems, Game Theory
- **Clubs:** Active member of **Databased** (CS Club); contributed to technical talks, demos, and coding events

Experience

Summer Research Intern, ML Lab, CSA, IISc — May 2025 – Present
Advisor: Prof. Chiranjib Bhattacharyya

Editing Concepts in Stable Diffusion [GitHub] — *July 2025*

- Developed a model editing approach for **Stable Diffusion** to unlearn specific concepts, as part of the **U&ME Challenge by ICCV 2025**.
- Achieved **20**% improvement in average forget score over diffusion-editing baselines in literature.
- **Technology:** Stable Diffusion, CLIP, PyTorch, Accelerate.

Diffusion Large Language Model [GitHub]: — *June 2025*

- Implemented Text Diffusion Model based on **Discrete Flow Matching** and trained it on the WikiText-103.
- Scaled training using **distributed data parallelism** across 4 GPUs to make training practical.
- **Technology:** Hugging Face Accelerate, PyTorch, GPT2.

Text-Conditioned Image Synthesis [GitHub] — *May 2025*

- Implemented a **Latent Diffusion Model** with **CLIP text encoder** for conditional image generation.
- **Technology:** PyTorch, Hugging Face Diffusers, Hugging Face Transformers

Projects

Diffusion Models [GitHub] — *March – April 2025*

- Studied influential papers on generative modeling, VAE, DDPM, DDIM, and Score-Based Generative Models.
- Implemented and benchmarked DDPM, DDIM, and Score-Based Models from scratch.
- **Technology:** PyTorch, Hugging Face Diffusers

Neural Network From Scratch [GitHub] — *June 2024*

- Built a minimal deep learning framework in NumPy, replicating core TensorFlow-style APIs.
- Implemented popular optimizers including SGD, Momentum, and Adam from scratch and benchmarked it on the MNIST dataset.

Easy 21 [GitHub] — *July 2024*

- Implemented and benchmarked classic Reinforcement Learning algorithms including Monte Carlo Control, SARSA, and Linear Function Approximation on Easy21, a simplified blackjack-like environment.

Skills

Programming Languages: Python, JavaScript, Bash

ML Libraries: PyTorch, Hugging Face (Transformers, Diffusers, Accelerate)

Data Analysis & Visualization: Pandas, Matplotlib

Tools & Platforms: Jupyter Notebook, Google Colab, Kaggle, Git, VS Code